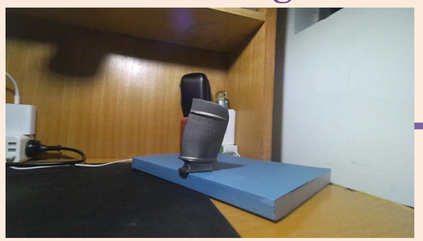
Three-Channel Condition

$$\mathbf{C} = \mathrm{Cat}_{\mathrm{ch}}\left(\mathbf{M}_{bal}, \tilde{\mathbf{G}}, \tilde{\mathbf{P}}\right) \in \mathbb{R}^{3\times H\times W}$$

RGB image



Fused Patch Embeddings

Transformer Backbone with Sparse Residual MoE

Token Sequence Construction

$$X_l = [\mathbf{x}_{\mathrm{CLS}}, \mathbf{t}_1, \ldots, \mathbf{t}_{1369}] \in \mathbb{R}^{1370\times 768}$$

1 global token + 1369 spatial tokens

Pre-Norm Self-Attention

Intermediate Tokens U_l

Independent sigmoid gates Reflective/Non-flat/Edge

Pretrained General FFN

Overlapping selections

Output Tokens X_{l+1}

Multi-Scale Decoding and Metric Depth

Depth Decoder

Scale De-Normalization

